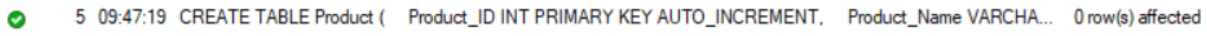
Seller & Inventory Management System

1. Create Tables

Seller

- Seller_ID
- Seller_Name
- Email
- Phone
- Address

```
CREATE TABLE Product (
   Product_ID INT PRIMARY KEY AUTO_INCREMENT,
   Product_Name VARCHAR(100) NOT NULL,
   Price DECIMAL(10,2) NOT NULL,
   CHECK (Price > 0)
);
```



Inventory

- Inventory_ID
- Product_ID
- Seller_ID
- Stock_Quantity
- Stock_Status
- Last_Updated

```
CREATE TABLE Inventory (
   Inventory_ID INT PRIMARY KEY AUTO_INCREMENT,
   Product_ID INT NOT NULL,
   Seller_ID INT NOT NULL,
   Stock_Quantity INT NOT NULL,
   Stock_Status VARCHAR(20) NOT NULL,
```

```
    Last_Updated DATE NOT NULL,

    CONSTRAINT fk_inventory_product
        FOREIGN KEY (Product_ID)
        REFERENCES Product(Product_ID),

    CONSTRAINT fk_inventory_seller
        FOREIGN KEY (Seller_ID)
        REFERENCES Seller(Seller_ID),

    CONSTRAINT check_stock_quantity
        CHECK (Stock_Quantity >= 0),

    CONSTRAINT check_stock_status
        CHECK (Stock_Status IN ('Available', 'Out of Stock'))
);
```



2. Insert Data

```
INSERT INTO Seller (Seller_Name, Email, Phone, Address)
VALUES
('ABC Traders', 'abc@gmail.com', '9876543210', 'Chennai'),
('Tech World', 'techworld@gmail.com', '9876543211', 'Coimbatore'),
('Smart Mart', 'smartmart@gmail.com', '9876543212', 'Madurai'),
('Digital Hub', 'digitalhub@gmail.com', '9876543213', 'Trichy'),
('Global Stores', 'globalstores@gmail.com', '9876543214', 'Salem');
SELECT * FROM Seller;
```

Seller_ID	Seller_Name	Email	Phone	Address
1	ABC Traders	abc@gmail.com	9876543210	Chennai
2	Tech World	techworld@gmail.com	9876543211	Coimbatore
3	Smart Mart	smartmart@gmail.com	9876543212	Madurai
4	Digital Hub	digitalhub@gmail.com	9876543213	Trichy
5	Global Stores	globalstores@gmail.com	9876543214	Salem
NULL	NULL	NULL	NULL	NULL

```
INSERT INTO Inventory
(Product_ID, Seller_ID, Stock_Quantity, Stock_Status, Last_Updated)
VALUES
(1, 1, 25, 'Available', '2026-09-28'),
(2, 1, 8, 'Available', '2026-09-28'),
(3, 2, 15, 'Available', '2026-09-28'),
(4, 2, 0, 'Out of Stock', '2026-09-28'),
(5, 3, 30, 'Available', '2026-09-28'),
(6, 3, 5, 'Available', '2026-09-28'),
(7, 4, 12, 'Available', '2026-09-28'),
(8, 5, 0, 'Out of Stock', '2026-09-28');
SELECT      * FROM Inventory;
```

Inventory_ID	Product_ID	Seller_ID	Stock_Quantity	Stock_Status	Last_Updated
17	1	1	25	Available	2026-09-28
18	2	1	8	Available	2026-09-28
19	3	2	15	Available	2026-09-28
20	4	2	0	Out of Stock	2026-09-28
21	5	3	30	Available	2026-09-28
22	6	3	5	Available	2026-09-28
23	7	4	12	Available	2026-09-28
24	8	5	0	Out of Stock	2026-09-28
NULL	NULL	NULL	NULL	NULL	NULL

2. Seller and Product Operaions

2.1 Display Products Supplied by Each Seller

```
SELECT
    s.Seller_Name,
    p.Product_Name,
    i.Stock_Quantity
FROM Seller s
JOIN Inventory i
    ON s.Seller_ID = i.Seller_ID
JOIN Product p
    ON i.Product_ID = p.Product_ID;
```

Seller_Name	Product_Name	Stock_Quantity
ABC Traders	Laptop	25
ABC Traders	Mouse	8
Tech World	Keyboard	15
Tech World	Monitor	0
Smart Mart	Printer	30
Smart Mart	Headphones	5
Digital Hub	Webcam	12
Global Stores	Speaker	0

2.2 Count Products Supplied by Each Seller

```
SELECT
    s.Seller_Name,
    COUNT(i.Product_ID) AS Product_Count
FROM Seller s
LEFT JOIN Inventory i
    ON s.Seller_ID = i.Seller_ID
GROUP BY s.Seller_ID, s.Seller_Name;
```

Seller_Name	Product_Count
ABC Traders	2
Tech World	2
Smart Mart	2
Digital Hub	1
Global Stores	1

2.3 Update Seller Details

```
UPDATE Seller
SET Address = 'Bangalore'
WHERE Seller_ID = 1;
```

Seller_ID	Seller_Name	Email	Phone	Address
1	ABC Traders	abc@gmail.com	9876543210	Bangalore
NULL	NULL	NULL	NULL	NULL

2. Seller and Product Operaions

2.1 Display Available Products

```
SELECT
   p.Product_ID,
   p.Product_Name,
   i.Stock_Quantity
FROM Product p
JOIN Inventory i
   ON p.Product_ID = i.Product_ID
WHERE i.Stock_Quantity > 0;
```

Product_ID	Product_Name	Stock_Quantity
2	Mouse	8
4	Monitor	0
6	Headphones	5
8	Speaker	0

2.2 Find Out-of-Stock Products

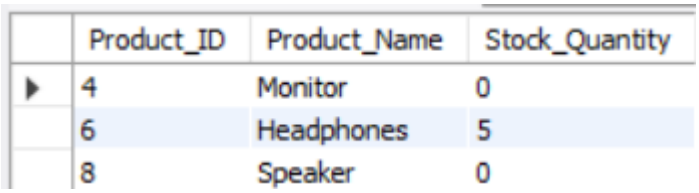
```
SELECT
   p.Product_ID,
   p.Product_Name
FROM Product p
JOIN Inventory i
   ON p.Product_ID = i.Product_ID
WHERE i.Stock_Quantity = 0;
```

Product_ID	Product_Name
4	Monitor
8	Speaker

2.3 Find Products with Stock Less Than 10

```
SELECT
   p.Product_ID,
   p.Product_Name,
   i.Stock_Quantity
FROM Product p
```

```
JOIN Inventory i
   ON p.Product_ID = i.Product_ID
WHERE i.Stock_Quantity < 10;
```



Product_ID	Product_Name	Stock_Quantity
4	Monitor	0
6	Headphones	5
8	Speaker	0

2.4 Update Stock Quantity

```
UPDATE Inventory
SET Stock_Quantity = 20,
   Stock_Status = 'Available',
   Last_Updated = CURDATE()
WHERE Product_ID = 2;
```

2.4 Update Stock Quantity

```
DELETE FROM Inventory
WHERE Stock_Quantity = 0;
```

3. Inventory Reports

3.1 Seller-wise product report

```
SELECT
   s.Seller_Name,
   p.Product_Name,
   i.Stock_Quantity,
   i.Stock_Status
FROM Seller s
JOIN Inventory i
   ON s.Seller_ID = i.Seller_ID
JOIN Product p
   ON i.Product_ID = p.Product_ID
ORDER BY s.Seller_Name;
```

Seller_Name	Product_Name	Stock_Quantity	Stock_Status
ABC Traders	Laptop	25	Available
ABC Traders	Mouse	20	Available
Digital Hub	Webcam	12	Available
Global Stores	Speaker	0	Out of Stock
Smart Mart	Printer	30	Available
Smart Mart	Headphones	5	Available
Tech World	Keyboard	15	Available
Tech World	Monitor	0	Out of Stock

3.2 Stock Availability Report

```
SELECT
    p.Product_Name,
    i.Stock_Quantity,
    i.Stock_Status,
    i.Last_Updated
FROM Product p
JOIN Inventory i
    ON p.Product_ID = i.Product_ID;
```

Product_Name	Stock_Quantity	Stock_Status	Last_Updated
Laptop	25	Available	2026-09-28
Mouse	20	Available	2026-09-28
Keyboard	15	Available	2026-09-28
Monitor	0	Out of Stock	2026-09-28
Printer	30	Available	2026-09-28
Headphones	5	Available	2026-09-28
Webcam	12	Available	2026-09-28
Speaker	0	Out of Stock	2026-09-28

3.3 Total Available Products

```
SELECT SUM(Stock_Quantity) AS Total_Available_Products
FROM Inventory
WHERE Stock_Quantity > 0;
```

	Total_Available_Products
▶	107

3.4 Total Out-of-Stock Products

```
SELECT COUNT(DISTINCT Product_ID) AS Out_of_Stock_Products
FROM Inventory
WHERE Stock_Quantity = 0;
```

	Category_Name	Product_Count

3.5 Highest Stocked Product

```
SELECT
   p.Product_Name,
   i.Stock_Quantity
FROM Product p
JOIN Inventory i
   ON p.Product_ID = i.Product_ID
WHERE i.Stock_Quantity = (
   SELECT MAX(Stock_Quantity)
   FROM Inventory
);
```

	Product_Name	Stock_Quantity
▶	Printer	30

3.5 Average Inventory Quantity

```
SELECT
   AVG(Stock_Quantity) AS Average_Inventory_Quantity
FROM Inventory;
```

	Average_Inventory_Quantity
▶	13.3750

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II BCA Batch I